



Affirmed Networks C-SGN Gateway Overview

Release 8.3.0.0

Updated: February 2019

Inside ...

- Overview of C-SGN
- CSGN Feature Descriptions

Affirmed Networks Inc.

35 Nagog Park
Acton, MA 01720
USA

Tel: +1-978-268-0800

www.affirmednetworks.com

Copyright© 2019 Affirmed Networks, Inc. All Rights Reserved.

AFFIRMED & DESIGN[®], ACUITAS SERVICE MANAGEMENT SYSTEM[®], AFFIRMED NETWORKS MOBILE CONTENT CLOUD[™], AFFIRMED OPEN WORKFLOW[™], AFFIRMED MOBILE CONTENT CLOUD[™] and other trademarks and designs are the registered or unregistered trademarks of Affirmed Networks, Inc. and its subsidiaries in the United States and in foreign countries. All other trademarks are the property of their respective owners. The Affirmed Networks, Inc. trademarks may not be used in connection with any product or service that is not Affirmed Networks' in any manner that is likely to cause confusion among customers or in any manner that disparages or discredits Affirmed Networks, Inc.

This document contains information that is the property of Affirmed Networks, Inc. This document may not be copied, reproduced, reduced to any electronic medium or machine readable form, or otherwise duplicated, and the information herein may not be used, disseminated or otherwise disclosed, except with the prior written consent of Affirmed Networks, Inc.

Affirmed NETWORKS, INC. SOFTWARE LICENSE AGREEMENT

AFFIRMED NETWORKS, INC. ("AFFIRMED") IS WILLING TO LICENSE THE ENCLOSED SOFTWARE AND ACCOMPANYING USER DOCUMENTATION (COLLECTIVELY, THE "PROGRAM") TO YOU ONLY UPON THE CONDITION THAT YOU ACCEPT ALL OF THE TERMS AND CONDITIONS OF THIS LICENSE AGREEMENT. PLEASE READ THE TERMS AND CONDITIONS OF THIS LICENSE AGREEMENT CAREFULLY BEFORE OPENING THE PACKAGE (S) OR USING THE AFFIRMED PRODUCTS CONTAINING THE SOFTWARE, AND BEFORE USING THE ACCOMPANYING USER DOCUMENTATION. OPENING THE PACKAGE (S) OR USING THE AFFIRMED PRODUCTS CONTAINING THE PROGRAM WILL INDICATE YOUR ACCEPTANCE OF THE TERMS OF THIS LICENSE AGREEMENT. IF YOU ARE NOT WILLING TO BE BOUND BY THE TERMS OF THIS LICENSE AGREEMENT, AFFIRMED IS UNWILLING TO LICENSE THE PROGRAM TO YOU, IN WHICH EVENT YOU SHOULD RETURN THE PROGRAM WITHIN TEN (10) DAYS FROM SHIPMENT TO THE PLACE FROM WHICH IT WAS ACQUIRED, AND YOUR LICENSE FEE WILL BE REFUNDED. THIS LICENSE AGREEMENT REPRESENTS THE ENTIRE AGREEMENT CONCERNING THE PROGRAM BETWEEN YOU AND AFFIRMED, AND IT SUPERSEDES ANY PRIOR PROPOSAL, REPRESENTATION OR UNDERSTANDING BETWEEN THE PARTIES.

License Grant

Subject to the provisions of this License, Affirmed grants to Licensee, and Licensee accepts, a non-exclusive, non-transferable license to use the object code form of the software supplied by Affirmed, including all patches, error corrections, updates and revisions thereto in machine-readable, object code form only (the "Software"), and the user documentation, including all updates and revisions thereto (the "User Documentation") for Licensee's internal business purposes (including, without limitation, in conjunction with Licensee's provision of services to its customers) on or in conjunction with the hardware product with which it was originally delivered or on a single designated computer. The Software and User Documentation are sometimes collectively referred to as the "Program". *Licensee acknowledges that certain Software provided by Affirmed may contain other restrictions and Licensee must refer to accompanying license certificates for each such Software licensed.* Licensee agrees that Licensee will not pledge, lease, rent, or share Licensee's rights under this License Agreement, and that Licensee will not, without Affirmed's prior written consent, assign or transfer Licensee's rights hereunder. Licensee agrees that Licensee may not duplicate (except as set forth below), decompile, disassemble, reverse engineer, modify, or otherwise translate the Software or permit a third party to do so and that Licensee shall not make the Program available to any other third party, or create works derivative of the Program, without Affirmed's express written consent. Licensee agrees it shall not publish in any fashion any results of benchmark tests run on the Program.

The Program is copyrighted and is only authorized to reproduce one copy of the Software and the User Documentation solely for backup purposes. Licensee is hereby prohibited from otherwise copying or translating, modifying or adapting the Program or, incorporating in whole or any part in any other product or creating derivative works based on all or any part of the Program. Licensee is not authorized to license others to reproduce any copies of the Program, except as expressly provided in this Agreement. Licensee agrees to ensure that all copyright, trademark and other proprietary notices of Affirmed affixed to or displayed on the Program will not be removed or modified. Licensee shall not decompile, disassemble or reverse engineer, the licensed software or any component thereof, except as may be permitted by applicable law, in which case Licensee must notify Affirmed in writing and Affirmed may provide review and assistance.

U.S. Government Restricted Rights. Notice - Distribution and use of products including computer programs and any related documentation and derivative works thereof, to and by the U.S. Government, are subject to the Restricted Rights provisions of FAR 52.227-19, paragraph (c)(2) as applicable, except for purchases by agencies of the Department of Defense (DOD). If the software is acquired under the terms of a Department of Defense or civilian agency contract, the software is a "commercial item" as that term is defined at 48 C.F.R. 2.101 (Oct. 1995), consisting of "commercial computer software" and "commercial computer software documentation" as such terms are used in 48 C.F.R. 12.212 of the

Software License Agreement

Federal Acquisition Regulations and its successors and 48 C.F.R. 227.7202-1 through 227.7202-4 (June 1995) of the DoD FAR Supplement and its successors. All U.S. Government end users acquire the software with only those rights set forth in this Agreement. Manufacturer is Affirmed Networks, Inc., 35 Nagog Park, Acton MA 01720. Unpublished - rights reserved under the copyright laws of the United States.

Affirmed's Rights

Licensee agrees that the Software and the User Documentation are proprietary, confidential products of Affirmed or Affirmed's licensor protected under U.S. copyright law and Licensee will use Licensee's best efforts to maintain their confidentiality. Licensee further acknowledges and agrees that all right, title and interest in and to the Program, including associated intellectual property rights, are and shall remain with Affirmed or Affirmed's licensor. This License Agreement does not convey to Licensee an interest in or to the Program, but only a limited right of use revocable in accordance with the terms of this License Agreement.

License Fees

The license fees paid by Licensee are paid in consideration of the license granted under this License Agreement.

Term

This License Agreement is effective upon Licensee's execution of this License Agreement and payment of the license fees and shall continue until terminated. Licensee may terminate this License Agreement at any time by returning the Program and all copies or portions thereof to Affirmed. Affirmed may terminate this License Agreement upon the breach by Licensee of any term hereof. Upon such termination by Affirmed, Licensee agrees to return to Affirmed the Program and all copies or portions thereof. Termination of this License Agreement shall not prejudice Affirmed's rights to damages or any other available remedy.

Limited Warranty

Affirmed warrants that with normal use and service each Software product shall materially conform to Affirmed's then current published specifications for the most current release of such Software product for a period of ninety (90) days from the date of shipment by Affirmed. Program media is warranted for ninety- (90) days from the date of shipment by Affirmed. Software support beyond these periods is available at additional cost under the terms of Affirmed's Annual Maintenance Service Agreement. During the warranty period, as Licensee's sole and exclusive remedy, Affirmed will correct a Software product's failure to conform to the warranty provided that Licensee has notified Affirmed in writing of the nature of the non-conformity. This warranty shall not apply if any Software product has been (i) modified or altered by anyone other than Affirmed, (ii) abused or misapplied, or (iii) used in combination with hardware or software other than the Affirmed manufactured products for which it was designed. Affirmed shall incur no liability under this warranty if Affirmed's tests disclose that the alleged defect is due to causes not within Affirmed's reasonable control, including alteration or abuse of the goods. If a Program is determined not to be defective or to have a defect due to causes not within Affirmed's reasonable control, Licensee agrees to pay for such repair at the repair price as listed in Affirmed's then current applicable price list. In no event does Affirmed warrant that the use of Software products will be error free or uninterrupted. Affirmed's sole obligation under the Software warranty shall be to provide the remedies described above.

EXCEPT FOR THE EXPRESS WARRANTIES STATED IN THIS WARRANTY, THE PROGRAM IS LICENSED "AS IS", AND Affirmed DISCLAIMS ANY AND ALL OTHER WARRANTIES, WHETHER EXPRESS, IMPLIED OR STATUTORY WITH RESPECT TO THE SOFTWARE PROVIDED UNDER THIS AGREEMENT, INCLUDING, BUT NOT LIMITED TO, ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE, ANY WARRANTIES OF NONINFRINGEMENT AS WELL AS ANY WARRANTIES ARISING FROM COURSE OF DEALING, USAGE OR TRADE PRACTICE.

Limitation of Liability

Affirmed's cumulative liability to Licensee or any other party for any loss or damages resulting from any claims, demands, or actions arising out of or relating to this License Agreement shall not exceed the greater of: (i) ten thousand U.S. dollars (\$10,000) or (ii) the total license fee paid to Affirmed for the use of the Program. In no event shall Affirmed be liable for any indirect, incidental, consequential, special, punitive or exemplary damages or lost profits or the loss of software or data, even if Affirmed has been

advised of the possibility of such damages. In the event that the Program contains the intellectual property of a third party pursuant to a license in favor of Affirmed, with a right to sublicense to Licensee, Affirmed hereby disclaims, to the extent permitted by law, such third party's liability of any damages, whether direct, indirect, incidental or consequential, arising out the use of the Program

Export Control

Licensee shall not export or re-export, directly or indirectly, any of the Programs (including any part of a Program or any direct product of such Program), Affirmed's Confidential Information or related technical data (a) into, or to a national resident of, any of those countries listed at the time of any shipment of the Products in the applicable U.S. export regulations as "prohibited or restricted" countries, or any other country to which such exports or re-exports may be restricted (collectively, the "Prohibited Countries") or (b) to anyone on the U.S. Treasury Department's list of Specially Designated Nations or the U.S. Commerce Department's Table of Denial Orders. At Licensee's request, Affirmed shall provide a list of the Prohibited Countries, Specially Designated Nations and/or Table of Denial Orders in order to assist Licensee to comply with such requirement.

Governing Law

This License Agreement shall be construed and governed in accordance with the laws and under the jurisdiction of the Commonwealth of Massachusetts, U.S.A. Any dispute arising out of this Agreement shall be referred to an arbitration proceeding in Boston, Massachusetts, U.S.A. by the American Arbitration Association.

Miscellaneous

If any action is brought by either party to this License Agreement against the other party regarding the subject matter hereof, the prevailing party shall be entitled to recover, in addition to any other relief granted, reasonable attorneys' fees and expenses of arbitration. Should any term of this License Agreement be declared void or unenforceable by any court of competent jurisdiction, such declaration shall have no effect on the remaining terms hereof. The failure of either party to enforce any rights granted hereunder or to take action against the other party in the event of any breach hereunder shall not be deemed a waiver by that party as to subsequent enforcement of rights or subsequent actions in the event of future breaches.

Affirmed has made no commitments or promises orally or in writing with respect to delivery of any future software features or functions. In relation to any future software features or functions, all presentations, RFP responses and/or product outlook documents, information or discussions, either prior to or following the date herein, are for informational purposes only, and Affirmed has no obligation to provide any future releases or upgrades or any features, enhancements or functions, unless specifically agreed to in writing by both parties. Reseller acknowledges that no purchasing decisions are based upon any future software features or functions.

**Third Party
Beneficiary**

If the Program incorporates or otherwise contains any intellectual Property of any third party pursuant to a license agreement in favor of Affirmed and sublicensed to Licensee, such third party shall, to the extent permitted by law, be a third party beneficiary of the terms and conditions of this Agreement.

Contents

Preface

Audience	ix
How to Use This Guide	x
Conventions	x
Related Documentation	xi
How to Obtain Product Documentation	xii
Technical Assistance Center	xii

Chapter 1 Overview of C-SGN function

EPS Network Architecture	1-2
MME Interfaces	1-2
S-GW Interfaces	1-3
P-GW Interfaces	1-3
Optimized EPS network architecture for CIoT	1-4
VM Types/Deployment Models for C-SGN function	1-6
VM Types and Personalities	1-7
VM footprint requirements	1-7
Capacity and Performance	1-9

Chapter 2 Features of C-SGN function

Enable/disable TACACS AAA for a CALEA group	2-1
Increase downlink data buffer size on the SGW	2-1
Support for mixed length IMEI/ IMEISV	2-2
Support for Legal Intercept logging function	2-2
Rejection of signaling for a group of UEs based on IMSI or LAPI	2-4
Configuration	2-4
Limitations	2-6
Use Case	2-6
Statistics	2-7
CDRs for NB-IoT related AVPs	2-8
Gy Interface	2-8
Gx Interface	2-9
Gz/Rf Interface PGW AVPs	2-9
Gz/Rf Interface SGW AVPs	2-9
Gz/Rf and Ga (GTPP) Interface SGW Event Trigger	2-10

Contents

PGW local CDRs (CSV and ASN.1) and GTPP CDRs (Ga Interface)	2-10
SGW local CDRs (CSV and ASN.1) and GTPP CDRs (Ga interface)	2-11
Support for RAT handover between EUTRAN and NBIOT	2-12
Support for inter-handoff/SGW	2-12
Support for a local database to increase PGW capacity	2-13
Limitations	2-13
CLI Commands	2-14
Performance Statistics	2-16
Show, Clear, and Count Commands	2-17
Procedure-duration Statistics	2-18
eDRX control message rejection handling at PGW	2-19
Configuration	2-19
Show Commands	2-20
Logs	2-21
Limitations	2-22
P-GW Congestion Protection for MTC use cases (SM-Back-off)	2-23
Support for SGW extended buffering	2-24
Support for non-IP data via tunneling (non-IP PDN type)	2-25
Protocol configuration options for MTU size of non-IP data over NAS	2-26
NBIOT PGW support for new PDN type	2-27
Support for the NBIOT RAT Type	2-28
Added support for ip and nonip PDN types.	2-29
CLI command to allow account administrators to change the default SFTP directory of user accounts	2-29

Preface

This guide describes an overview of the Affirmed Networks CIoT Serving Gateway Node (C-SGN) service running on the Mobile Content Cloud (MCC).

This guide also describes the features specific to the C-SGN Gateway function, CLI objects, and commands used to configure these features.

For detailed information on CLI commands, CDR, lawful intercept, performance statistics and KPIs, refer to the documents listed in [Related Documentation \(page -xi\)](#).

***Note:** The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation team staff, and other qualified service personnel responsible for provisioning and monitoring the system. The audience should be familiar with Long Term Evolution (LTE), 3G, or 4G networks.

How to Use This Guide

This guide contains the following information:

Chapter/Appendix	Describes
Chapter 1, Overview of C-SGN function	Provides a high-level overview of the Affirmed Networks C-SGN Gateway node
Chapter 2, Features of C-SGN function	Describes the features available in the current release. For each feature, related CDRs, Configuration Commands, and Statistics are described as applicable.

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<courier bold italic blue font>	configure_dr convert slave --masterhost <host IP>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.	 	Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<host name>
Asterisk after a CLI object indicates a required object.	*	name*

Variables, book and chapter titles, file path names, new terms, and emphasized text

Times New Roman italic font

See the *Acuitas User's Guide*.

Tip: Helpful suggestions.

Note: *Additional information that may apply to the subject text.*



CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

Related Documentation

This document describes the C-SGN features available in the current release. For each feature, the related CDRs, Configuration Commands, and Statistics are described as applicable.

For detailed information about CDRs, configuration CLI commands, and statistics, refer to the following documents:

- *Affirmed Networks Gateway Administration Guide, Release 8.3.0.0*
- *Affirmed Networks Charging Data Records Specification, Release 8.3.0.0*
- *Affirmed Networks CLI Reference Guide, Release 8.3.0.0*
- *Affirmed Networks Lawful Intercept Module Specification and Provisioning Guide, Release 8.3.0.0*
- *Acuitas Performance Statistics, Release 8.2.0.0*
- *Acuitas User's Guide, Release 8.2.0.0*
- *Affirmed Networks C-SGN Documentation Summary*

For information about MCC and MME features and their configuration, refer to the respective MCC and MME documentation.

Refer to the *Affirmed Networks C-SGN Documentation Summary* document for a complete list of documentation available for C-SGN.

How to Obtain Product Documentation

Documentation is available for currently supported product releases. Documentation is available online in Adobe PDF format. You can view PDFs online using the Adobe Reader[®]. To download the latest version of the Adobe Reader software from the Adobe web site, go to <http://get.adobe.com/reader/>

Product documentation is available from the Affirmed Networks SFTP server. Contact your sales representative or the TAC for instructions on accessing the SFTP server.

Technical Assistance Center

Affirmed Networks Technical Assistance Center (TAC) provides technical support 24 hours a day, 7 days a week (24 x 7) for any customer with an active maintenance contract.

To contact the TAC:



E-mail: prodsupport@affirmednetworks.com



24 x 7 Telephone support:

Local: 1-978-268-0871

Toll free: 1-888-283-2838

Overview of C-SGN function

The Affirmed Networks Cellular Internet of Things (CIoT) Serving Gateway Node (C-SGN) is a fully virtualized EPC node that is optimized to support the C-IOT network. It is a combined EPC node that collocates EPS entities in the control and user planes (MME, S-GW, P-GW). The external C-SGN interfaces are the respective EPC entities supported by the C-SGN, such as the MME, S-GW, and P-GW.

***Note:** Scope of this document is limited to an overview of the Affirmed C-SGN function (combined P-GW + S-GW) and its features. For information on MME and its features, refer to the Affirmed Networks MME documentation.*

Operators can deploy the C-SGN in the following configurations:

- Combined MME, S-GW, and P-GW
- Combined MME and S-GW. The C-SGN works with an external PGW for small data communication between the CIoT devices and their corresponding application servers.

On top of the existing EPC capabilities of the C-SGN, it also provides and supports the following optimization for the CIoT devices:

- Narrow band eNodeBs. The C-SGN connects to the NB-eNodeB via the S1-lite interface.
- Control plane CIoT EPS optimization for small data transmission. The C-SGN supports small data transmission using the NAS signaling layer. This is also referred to as DoNAS (Data over Non Access Stratum).
- Security procedures for efficient small data transmission. The small data transmission is integrity protected and encrypted.
- SMS without combined attach for NB-IoT only UEs. The CIoT device can perform SMS without performing combined attach procedure.

- Non-IP data transmission via SGi tunneling. Added support for new PDN type used to deliver non-IP data.
- Attach without PDN connectivity. The UE can attach without activating a PDN session.
- Power savings. The C-SGN supports many features to allow the CIoT devices to conserve battery, for example, Power Saving Mode, Extended DRX, Paging Optimization, etc.

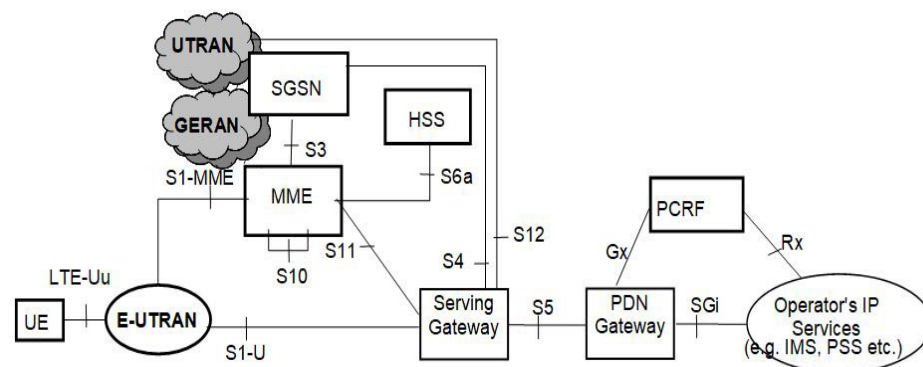
EPS Network Architecture

The CIoT network evolves from the existing EPS network. This section gives a brief introduction of the EPS network.

The EPS network is an all-IP network, which is a major deviation from the previous generations of wireless packet data network. This section primes the relevant EPC nodes that are part of the C-SGN, namely the MME, the S-GW and the P-GW.

The following figures show the network architecture as defined by TS 23.401.

Figure 1-1 Non-roaming architecture for 3GPP access



MME Interfaces

The following interfaces are related to the MME:

- **S1-MME** - Interface used to connect the E-UTRAN access nodes to the MME for signaling. The data plane interface is the S1-U between the access nodes to the Serving Gateway.
- **S3** - Interface between the MME and the S4-SGSN for mobility management purpose.
- **S6a** - Interface between the MME and the HSS for subscription and authentication information management.

- S10 - Interface between two MME nodes for inter-MME mobility management.
- S11 - Interface between the MME and the S-GW for PDN and bearer management. The S-GW extends the signaling from the MME to the P-GW via the S5 or S8 interface.

S-GW Interfaces

The following interfaces are related to the S-GW:

- S1-U - Interface used to carry user bearer packets between the E-UTRAN access nodes and the core network.
- S4 - Interface between the SGSN and the S-GW. This interface carries both GTP-C signaling packets as well GTP-U user bearer packets.
- S11 - Interface between the MME and the S-GW as discussed above.
- S12 - Interface between the UTRAN access nodes and the S-GW. This is similar to the S1-U interface in the E-UTRAN.
- S5/S8 - Interface between the S-GW and the P-GW. This interface carries both GTP-C signaling packets and GTP-U user bearer packets. The S5 interface is used between an S-GW and a P-GW that belong to the same PLMN, whereas the S8 interface is between an S-GW and a P-GW that belong to different PLMNs.

P-GW Interfaces

The following interfaces are related to the P-GW:

- S5/S8 - Interface between the S-GW and the P-GW as discussed above.
- Sgi - Interface between the P-GW and the Operator's IP Services network. This interface carries regular IP packets as in an IP network.
- Gx - Interface between the P-GW and the PCRF for policy management.

The readers are referred to TS23.401 for more details on entities and interfaces mentioned in the figures above.

Note: Please note, EPC network architecture figures in this section do not cover all the interfaces used in the network. For example, the SGs interface between the MME and the VLR/MSC. This interface allows UE in the 4G network to fall back to 2G GSM or 3G UMTS network for voice call. Another interface not depicted here is the S13 interface between the MME and the EIR for equipment validation.

Optimized EPS network architecture for CIoT

The EPS optimized for CIoT is intended to serve UEs that are of vastly different characteristics as compared to typical consumer devices. The CIoT devices are of ultra-low power consumption so as to last a long lifespan. The density of devices is expected to be much higher compared to the regular EPS network. The CIoT network also uses narrow band radio access technology as data rate per device is very low.

An EPS optimized for CIoT can be enabled by having sub-set of functionalities implemented in single logical entity C-SGN (CIoT Serving Gateway Node). The C-SGN thus can contain the MME, S-GW and P-GW with specific functionalities that are required by CIoT.

Figure 1-2 and Figure 1-3 below illustrates the C-SGN based network architecture and the high level interface view of the C-SGN in the non-roaming and roaming scenarios. Mobility and Attach procedures are performed the same way as regular EPS network with corresponding entities MME, S-GW and P-GW.

Figure 1-2 C-SGN interfaces (non-roaming)

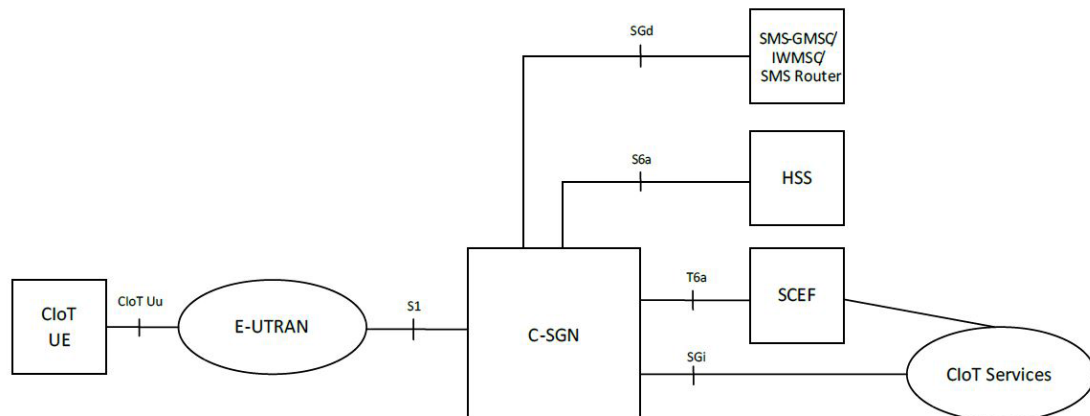
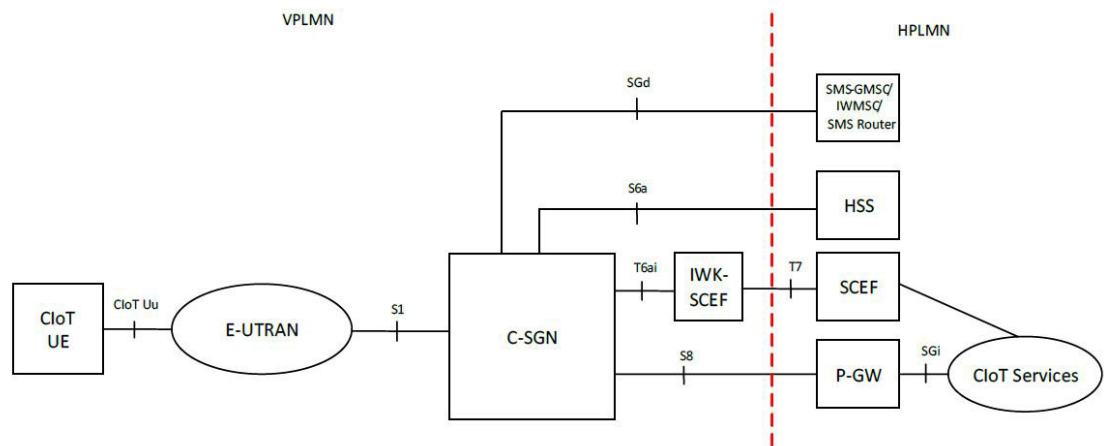


Figure 1-3 C-SGN interfaces (roaming)



The C-SGN (CIoT Serving Gateway Node) is a combined node EPC implementation option that minimizes the number of entities by collocating EPS entities in the control and user planes paths (e.g.MME, S-GW, P-GW), which may be preferred in CIoT deployments. The external interfaces of CSGN implementation option are the interfaces of the respective EPC entity supported by the C-SGN, such as MME, S-GW, and P-GW.

As depicted in the figures above, the following new interfaces are introduced to support the CIoT optimization:

- SGd - This interface is used between the MME within the C-SGN and the SMS-SC. It allows SMS to be delivered to CIoT devices via the MME.
- T6a and T6ai - This interface is used between the MME within the C-SGN and the SCEF or interworking SCEF in the roaming case. This provides the ability to support Non-IP Data Delivery (NIDD) for CIoT devices.

In [Figure 1-2](#), note that the C-SGN has an SGi interface directly – this implies that the PGW functionality is subsumed in the logical C-SGN function. Internally, the logical C-SGN is composed of the MME, SGW and (optionally) PGW.

The readers are referred to TS23.401 and TS23.682 on CIoT and Machine Type Communications.

VM Types/Deployment Models for C-SGN function

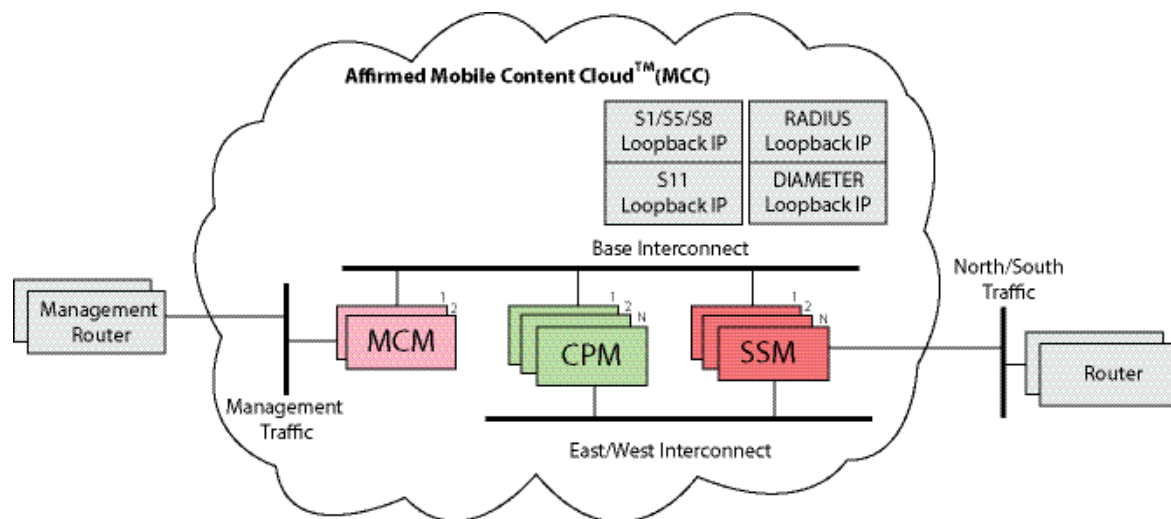
In order to achieve a smaller overall footprint when dynamic scale is not required, Affirmed Networks supports the following VM combination types to facilitate a smaller overall VM count within the cluster.

A minimum of three VM types is required to support full cluster capability:

- vMCM
- vCPM
- vSSM

Figure 1-4 below provides a high-level view of the cluster interconnect topology for a small deployment model with basic redundancy.

Figure 1-4 Small Deployment Model



VM Types and Personalities

C-SGN function requires the following three VM types that can be specialized into personalities for larger deployments:

VM Type	Description
Management Configuration Module (vMCM)	■ A Management Configuration Module VM instance running core processes, including node and configuration functions ■ Two vMCMs are required per cluster for redundancy
Control Plane Module (vCPM)	■ vCPM runs all tasks needed for call-control functionality. ■ Two vCPM are required per for redundancy
Subscriber Services Module (vSSM)	■ A Subscriber Services Module VM instance running Datapath processes and supporting infrastructure services ■ vSSM is responsible for receiving/transmitting packets to/from MCC and workflow services

VM footprint requirements

The following tables lists the VM sizing requirements to support 1.5M Simultaneous Attached Subscribers (SAU) and 500 transactions per second.

The P-GW and S-GW can be on separate nodes or combined into a single node called SAEGW.

Table 1-1 Footprint required for separate SGW and PGW nodes

	SGW			PGW		
VM Type	vCPU	Memory (GB)	Disk (GB)	vCPU	Memory (GB)	Disk (GB)
MCM1	4	16	520	6	16	520
SSM1	6	48	82	8	64	82
CPM1	6	48	82	8	64	82
MCM2	4	16	520	6	16	520
SSM2	6	48	82	8	64	82
CPM2	6	48	82	8	64	82
Total	32	224	1368	44	288	1368

Table 1-2 Footprint required for SAEGW (SGW + PGW combined node)

	SAEGW (PGW+SGW)		
VM Type	vCPU	Memory (GB)	Disk (GB)
MCM1	4	16	520
SSM1	8*	64*	82
CPM1	8	64	82
MCM2	4	16	520
SSM2	8*	64*	82
CPM2	8	64	82
Total	40	288	1368

* vCPUs and Memory can be adjusted across VMs based on performance tuning

Table 1-3 provides an illustrative sample of the call model supported by the VM footprint listed above. The table lists number of events per subscriber, per hour, for a normal hour.

Table 1-3 Call Model supported with rate of events

Type of Events	# of Events per hour per subscriber
Attach Request (with PDN)	0.001
Attach Request (without PDN)	0.001
Detach	0.002
Tracking Area Update	0.05
Inter MME Tracking Area Update	0
Paging	0.1
Control Plane Service Request	0.5
S1 UE context release (active to idle)	0.5
MO SMS	0.1
MT SMS	0.1

Capacity and Performance

Table 1-1 and Table 1-2 above lists the VM footprint required to achieve 1.5 million simultaneous sessions and 500 transactions per second (TPS) with a configuration of two MCM, two CPM, and two SSM VMs. Additional CPM and SSM VMs may be added to scale up the capacity and performance. Each additional pair of CPM and SSM VMs can support additional 1.5M sessions and 500 TPS.

Features of C-SGN function

This chapter describes the features and enhancements of the C-SGN function (S-GW + P-GW) included in Release 8.3.0.0.

Enable/disable TACACS AAA for a CALEA group

Support for CALEA configuration option to enable/disable TACACS AAA for a CALEA group.

Using the default calea account, issue the following command to enable/disable the feature:

```
an3000-mcm-slot7cpu1(config)# calea tacacs-plus state [enabled | disabled]
```

When the tacacs-plus feature is enabled (default), when a user logs in, Lawful Intercept sends an authentication request to the TACACS+ server. Upon successful authentication, the server responds with the user's group assignment.

When the tacacs-plus feature is disabled for a CALEA group, the user is not authenticated by the TACACS+ server. User login and all commands entered are not validated.

Increase downlink data buffer size on the SGW

When the subscriber is in Idle mode, the SGW buffers the downlink data for later transmission to the subscriber. To accommodate subscribers who remain in an Idle state for a longer duration in Extended Paging mode, the SGW now uses the increased maximum Extended Paging downlink buffer size of 100 packets:

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway sgw <name>
paging-profile maxcap-downlink-pkts-to-buffer ?
```

Possible completions:

```
<unsignedInt, 1 .. 100>[5]
```

Support for mixed length IMEI/ IMEISV

MCC gateway performs Lawful Intercept based on a match of the first 14 digits of the IMEI/IMEISV received in GTP signaling.

IMEI (International Mobile Equipment Identity) is a unique number that identifies a device in the mobile network. GTP signaling may contain either the 15-digit IMEI or 16-digit IMEISV:

- The 15-digit IMEI includes a Type Allocation Code (TAC) of 8 decimal digits and a Serial Number (SNR) of 6 decimal digits plus a Spare decimal digit (which may be used to send a checksum value). When an LI target is provisioned for IMEI-based intercepts, the MCC gateway ignores the Spare Digit/Check Digit and performs Lawful Intercept based on a match of the 14 digits of TAC plus SNR.
- The 16-digit IMEISV includes the TAC and SNR plus a 2-decimal digit Software Version Number (SVN). When an LI target is provisioned for IMEI-based intercept, the MCC gateway ignores the Software Version Number field (SVN) and performs Lawful Intercept based on a match of the 14 digits of TAC plus SNR.

When provisioning a target for IMEI-based Lawful Intercept, the target's identityVal field (identity-value) may be provisioned with any of the following values:

- The first 14 digits of IMEI on which to perform the intercept (TAC + SNR)
- The entire 15-digit IMEI (TAC + SNR + Spare Digit/Check Digit)
- The complete 16-digit IMEISV (TAC + SNR + Software Version)

In each of these cases, the MCC gateway performs Lawful Intercept on a match of the first 14 digits of the provisioned identity-value against the first 14 digits of the IMEI received in the GTP signaling. The MCC gateway ignores the 15th and 16th digits (if present in the provisioned identity-value and/or GTP signaling) when checking for a match of the target identity-value with the subscriber's IMEI.

Support for Legal Intercept logging function

Added support to log CALEA user commands related to configure/modify/display of LI targets to a CALEA audit file. The logs can be output directly, such as using syslog, snmp, or sftp to a secure logging server. Logging is supported for both persistent and non-persistent targets.

Using the default CALEA account, issue the following command to enable the feature:

```
an3000-mcm-slot7cpu1(config)# infrastructure logging event-log policy  
liaudit admin-state enabled
```


When the `liaudit` feature is enabled, MCC configures the following default logging policy:

```
sinks liaudit
  type          file
  file-name      liaudit
  file-format    ascii
  display-field  displaystring
```

The CALEA user cannot create or delete these logging filters, but can modify `file-management profile` parameters to set different file transfer policies and set username/password. For example:

```
infrastructure file-management profile eventlog liaudit
  purge-thr          100
  purge-low-thr       0
  file-rotate-size-bytes 10485760
  file-transfer-enabled disabled
  upload-frequency    1hour
```

The CALEA account can also issue the `trigger-file-rotate` and `trigger-file-transfer` commands.

The audit log uses the following file format:

```
2017-10-09T03:00:06.789917543-0400 <userid> <LIID> <operation>
<result> [<attributename> <attributeValue>]
```

For example:

```
2017-11-24T05:23:59.282318654-0500 Sev=4 [CALEA] [CALEATarget] [ID=00]
[00012c0700011fe4] [28407 28724] [CaleaDirector_CaleaDirector_0]
"[CaleaUserName:calea Liid:mobster OperationType:createTarget
Operationresult:success]: dataCollectionType: iri-plus-cc gwType:
all-gw wlangwtype: none x2InterfaceName: X2-AN-V4 x3InterfaceName:
X3-AN-V4"

2017-11-24T05:24:21.029159989-0500 Sev=4 [CALEA] [CALEATarget] [ID=00]
[00012c0700011fe4] [28407 28724] [CaleaDirector_CaleaDirector_0]
"[CaleaUserName:calea Liid:NA OperationType:target-status
```

Rejection of signaling for a group of UEs based on IMSI or LAPI

CSGN supports a configurable option to reject signaling from a certain group on UEs based on IMSI. Operators configure a parameter to reject signaling from a set of UEs/USIM based on the IMSI range for a congested APN, when the load reaches beyond a certain configured threshold.

In addition, CSGN provides an option to configure a parameter to reject signaling from LAPI (Low Access Priority Indication) devices. Operators configure this feature using the `signaling-priority-lapi` key in subscriber-analyzer.

This feature is not limited to only IMSI (or) LAPI-based differentiation; it can also be used with other subscriber-analyzer rules as long as the desired workflow control-profile is selected. For example, imei-range, rat-type, and pdn-type etc.

The configured IMSI range or LAPI is associated with a critical, major, or minor threshold using `workflow control-profile congestion-reject-level`.

This feature also provides congestion control action enforcement for remapped APNs. CSGN applies the congestion control action on the final remapped APN, after choosing the subscriber-analyzer and workflow control-profile.

This feature supports:

- A new `subscriber-analyzer signaling-priority-lapi` key
- Configurable APN threshold congestion rejection level in workflow control-profile
- Decoding and encoding signaling priority indication
- Decoding of LAPI received in signaling priority indication IE in create-session-request message
- PGW and APN-level counters for create-session-request attempt, accept, and reject with LAPI bit set in signaling priority indication IE
- Reject create-session-request (or new sessions) at PGW, if selected final APN is in congestion state
- Cause code **APN Congestion (113)** and **PGW Back-off Time** in create-session-response if the new session is rejected due to APN congestion

Configuration

Configure `signaling-priority-lapi` under subscriber-analyzer key. This key is used to define criteria for the subscriber-analyzer selection and configure rules to differentiate the treatment for LAPI PDNs.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer ANALYZE1
an3000-mcm-slot7cpu1(config-subscriber-analyzer-ANALYZE1)#
an3000-mcm-slot7cpu1(config-subscriber-analyzer-ANALYZE1)# key test
an3000-mcm-slot7cpu1(config-key-test)#
an3000-mcm-slot7cpu1(config-key-test)# signaling-priority-lapi ?
Possible completions:
    <true/false>
an3000-mcm-slot7cpu1(config-key-test)# signaling-priority-lapi true
an3000-mcm-slot7cpu1(config-key-test)# commit
Commit complete.
an3000-mcm-slot7cpu1(config-key-test)#
```

Enable the **apn-congestion-handling** feature:

```
an3000-mcm-slot7cpu1(config)# zone default
an3000-mcm-slot7cpu1(config-zone-default)# gateway apn consumer.att
an3000-mcm-slot7cpu1(config-apn-consumer.att)# apn-congestion-handling
```

Enable the **admin-state** and configure apn-congestion-threshold limits:

```
an3000-mcm-slot7cpu1(config-apn-consumer.att)#
apn-congestion-threshold ?
Possible completions:
    admin-state      Enable/Disable the administrative state of this
threshold
    critical-enter    Crossing critical threshold
    major-enter       Crossing major threshold
    minor-enter       Crossing minor threshold
    threshold-type
an3000-mcm-slot7cpu1(config-apn-consumer.att)#
apn-congestion-threshold admin-state enabled critical-enter 10

an3000-mcm-slot7cpu1(config-apn-consumer.att)# commit
Commit complete.
```

Configure congestion back off **timer-unit** value. This configuration causes the gateway to reject the call with a cause code of **113 EPC timer with configured value** if APN is congested. The system detects congestion if the number of calls is greater than or equal to the configured critical threshold. In the example below, if the number of calls ≥ 10 and the number of calls falls below 10, then the gateway starts accepting new calls for that APN. Use the **admin-state** to control the configuration instead of removing and re-adding the configuration. The **threshold-type** is always set to **op** and has no impact on feature behavior.

```
apn-congestion-timer critical-enter timer-unit <timer-enum>
apn-congestion-timer critical-enter timer-value <1-31>
```

Define the **apn-congestion-reject-level** to reject new sessions. The workflow control-profile congestion-rejection-level takes priority over the apn-congestion-reject-level.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile CTRLPROF
an3000-mcm-slot7cpu1(config-control-profile-CTRLPROF)#
an3000-mcm-slot7cpu1(config-control-profile-CTRLPROF)#apn-congestion-reject-level ?
```

Possible completions:

```
critical      Start rejecting new sessions on crossing critical
threshold
major         Start rejecting new sessions on crossing major threshold
minor         Start rejecting new sessions on crossing minor threshold
none          No apn-congestion reject
an3000-mcm-slot7cpu1(config-control-profile-CTRLPROF)#
apn-congestion-reject-level major-enter
an3000-mcm-slot7cpu1(config-control-profile-CTRLPROF)#commit
Commit complete
```

Limitations

The following limitations apply to this feature:

- CSGN enforces the apn-congestion-reject action on the final APN after running subscriber-analyzer. The final APN can be original or remapped. In the remapped APN case, the original APN is passed even if it is congested, if the final APN is not in a congested state. Sessions are counted as part of the final APN.
- CSGN does not decode LAPI in bearer resource command, therefore, for the existing session, after the APN enters into congestion, if a new dedicated bearer activation arrives, the CSGN does not enforce the apn-reject-congestion-level action.
- Low priority access inclusion in CDRs is not supported.
- Gateway performance statistics for a number of current LAPI PDN sessions are not supported.
- Alarms relating to APN congestion are not supported.

Use Case

If the PGW APN is congested, it rejects the new calls for a certain group of UEs located on a specific APN. Also, the PGW continues to send the back-off timer to the MME, therefore, the same call is redirected to another PGW or retried after the congestion-back-off-timer expires.

The following use case applies where the Operator wants to reject new sessions (or PDNs) for some IMSI range (or LAPI devices) at a minor congestion level, and another set of UE's sessions (or PDNs) at critical level. To realize this use case:

- 1 Configure critical and minor congestion levels at the APN level and enable the admin state. This step is mandatory.
- 2 Define the configuration such that the desired IMSI range or LAPI devices use a separate workflow control-profile and set the apn-congestion-reject level to minor.
- 3 For another set of UEs, select a separate workflow control-profile and set the apn-congestion-reject level to critical.

Note: If workflow control-profile congestion-control-reject-level is not defined, by default, the apn-congestion-reject-level enforcement occurs. All new sessions in this category are ejected if the APN is in congested state.

Statistics

This feature supports the following show commands and statistics:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance apn consumer.att
```

```
<<snip>>
```

```
create-session-rejects-apn-in-congestion-state: 0
```

```
<<snip>>
```

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw
```

```
<<snip>>
```

```
create-session-rejects-apn-in-congestion-state: 0
```

```
<<snip>>
```

```
an3000-mcm-slot7cpu1#
```

Displays the number of calls rejected in each state due to congestion:

```
show zone default gateway statistics cluster-level call-performance
apn <apn_name>
```

```
<<snip>>
```

```
create-session-rejects-apn-in-critical-state 0
```

```
create-session-rejects-apn-in-major-state 0
```

```
create-session-rejects-apn-in-minor-state 0
```

```
<<snip>>
```

Provides information about create-session-request received with low access priority.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance apn consumer.att
```

```
<<snip>>
```

```
create-session-lapi-pdn-attempts      0
create-session-lapi-pdn-accepts      0
create-session-lapi-pdn-rejects      0
<<snip>>
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw
<<snip>>
create-session-lapi-pdn-attempts      0
create-session-lapi-pdn-accepts      0
create-session-lapi-pdn-rejects      0
<<snip>>
an3000-mcm-slot7cpu1#
```

CDRs for NB-IoT related AVPs

Added support for NB-IoT related AVPs on the Gx, Gy, and Gz interfaces and added NB-IoT related SGW/PGW local and GTPP CDRs for use with the C-SGN.

Also see the *Affirmed Networks Charging Data Records Specification, Release 8.3.0.0*

Gy Interface

Added support for the following AVPs on the Gy interface for use with the C-SGN as defined in 3GPP TS 32.299 V13.9.0. These AVPs are configurable in CCR data record templates. The default `presentFlag` setting is `no`, indicating the PGW does not include the AVP in the CCR message.

- APN-Rate-Control
- SCS-AS-Address
- Serving-PLMN-Rate-Control
- SGi-PtP-Tunnelling-Method
- UNI-PDU-CP-Only-Flag
- 3GPP-PDN-Type – Added two new values:
 - 3 = Ipv4v6
 - 4 = Non-IP
- 3GPP-RAT-Type – Added new value: EUTRAN-NB-IoT (8)

Gx Interface

Added a new RAT type in the RAT Type AVP sent on the Gx interface in CCR messages. The C-SGN includes the new EUTRAN-NB-IoT (1005) RAT type in the RAT-Type AVP for NB-IoT PDNs.

Gz/Rf Interface PGW AVPs

Added support for the following PGW AVPs on the Gz/Rf interface for use with the C-SGN as defined in 3GPP TS 32.299 V13.9.0. These AVPs are configurable in ACR data record templates. The default `presentFlag` setting is `no`, indicating the PGW does not include the AVP in the ACR message.

- Service-Information → PS-Information
 - APN-Rate-Control
 - Low-Priority-Indicator
 - SCS-AS-Address
 - Serving-PLMN-Rate-Control
 - SGi-PtP-Tunnelling-Method
 - UNI-PDU-CP-Only-Flag
 - 3GPP-PDN-Type – Added two new values: 3 = Ipv4v6 and 4 = Non-IP
 - 3GPP-RAT-Type – Added new value: EUTRAN-NB-IoT (8)
- Service-Data-Container
 - APN-Rate-Control
 - 3GPP-RAT-Type
 - Serving-PLMN-Rate-Control

Gz/Rf Interface SGW AVPs

Added support for the following SGW AVPs on the Gz/Rf interface for use with the C-SGN as defined in 3GPP TS 32.299 V13.9.0. These AVPs are configurable in ACR data record templates. The default `presentFlag` setting is `no`, indicating the SGW does not include the AVP in the ACR message.

- Service-Information → PS-Information
 - CP-CIoT-EPS-Optimisation-Indicator
 - Low-Priority-Indicator
 - Serving-PLMN-Rate-Control
 - UNI-PDU-CP-Only-Flag
 - 3GPP-PDN-Type – Added two new values: 3 = Ipv4v6 and 4 = Non-IP
 - 3GPP-RAT-Type – Added new value: EUTRAN-NB-IoT (8)

- Traffic-Data-Volumes
 - CP-CIoT-EPS-Optimisation-Indicator
 - 3GPP-RAT-Type
 - Serving-PLMN-Rate-Control

Gz/Rf and Ga (GTPP) Interface SGW Event Trigger

Added a new event trigger to the event-trigger-template used to specify the events that trigger an interim update on the offline charging interface and to trigger the closure of an interim SGW CDR.

```
an3000-mcm-slot7cpu1(config)# service-construct
interface-trigger-template event-trigger-template <template-name>
change-in-up-to-ue-close-trigger
```

change-in-uptoue-close-trigger – Specifies whether a change in the User Plane (UP) to User Equipment (UE) (for example, a switch between S11-U and S1-U) triggers an update. Value 0 disables the trigger, value 1 - 20 enables the trigger. In the case of offline interfaces, the value specifies the maximum number of UP to UE Change events to trigger a CDR close.

Added a new change in charging condition to the Change-Condition AVP within the Service-Information → PS-Information AVP sent on the Gz/Rf interface.

36 = Change in UP to UE

Added the following change in charging conditions to the Change-Condition ASN.1 field within the Change of Service Condition field sent on the Ga interface.

```
userPlaneToUEChange (18)
servingPLMNRateControlChange (19)
```

PGW local CDRs (CSV and ASN.1) and GTPP CDRs (Ga Interface)

Added support for the following fields in PGW local CDRs (CSV and ASN.1) and GTPP CDRs for use with the C-SGN as defined in 3GPP TS 32.298 V13.7.0. These fields are configurable in local Start/Stop/Interim and GTPP Interim/Stop data record templates. The default `presentFlag` setting is `no`, indicating the field is not included in PGW CDRs.

- APN Rate Control
- Low Priority Indicator
- PDP/PDN Type Extension
- SCS/AS Address
- Serving PLMN Rate Control

- SGI PtP Tunnelling Method
- UNI PDU CP Only Flag
- 3GPP-RAT-Type – Added new value: EUTRAN-NB-IoT (8)
- List of Service Data (Change of Service Condition)
 - APN Rate Control
 - RAT Type
 - Serving PLMN Rate Control

Note: The List of Service and Traffic Data Volumes subfields are only configurable in local ASN.1 and GTPP Interim/Stop data record templates. They are not configurable in local CSV data record templates

SGW local CDRs (CSV and ASN.1) and GTPP CDRs (Ga interface)

Added support for the following fields in SGW local CDRs (CSV and ASN.1) and GTPP CDRs for use with the C-SGN as defined in 3GPP TS 32.298 V13.7.0. These fields are configurable in local Start/Stop/Interim and GTPP Interim/Stop data record templates. The default `presentFlag` setting is `no`, indicating the field is not included in SGW CDRs.

- CP CIoT EPS Optimisation Indicator
- Low Priority Indicator
- PDP/PDN Type Extension
- Serving PLMN Rate Control
- UNI PDU CP Only Flag
- 3GPP-RAT-Type – Added new value: EUTRAN-NB-IoT (8)
- List of Traffic Data Volumes (Change of Charging Condition)
 - CP CIoT EPS Optimisation Indicator
 - RAT Type
 - Serving PLMN Rate Control

Note: The List of Traffic Data Volumes subfields are only configurable in local ASN.1 and GTPP Interim/Stop data record templates. They are not configurable in local CSV data record templates.

Support for RAT handover between EUTRAN and NBIOT

This feature adds support for idle mode inter-RAT handover between the EUTRAN (Wideband-EUTRAN) and NB-IoT (Narrowband-EUTRAN).

This feature is useful in eliminating the need for a device to detach and reattach when moving between RATs with associated signaling and power consumption costs and the potential for incorrect RAT type information within the network.

The following use cases apply to this feature:

- WB-E-UTRAN to NB-IoT handoff – The NB-IoT capability can be incorporated into a smart phone or other WB-E-UTRAN device in order to locate a lost device or contact a user who is outside the WB-E-UTRAN coverage area. In addition, the TAU signaling from the UE to the network can be used to trigger events.
- WB-E-UTRAN to NB-IoT mobility – This feature can be used to return a high-performance device to the higher performance radio of WB-E-UTRAN.

Support for inter-handoff/SGW

This feature adds support for connected/idle mode handovers for CIoT features on CSGN within the same RAT type, that is NB-IoT or WB-E-UTRAN. These features include: non-IP PDNs, control plane CIoT optimization, switch between control plane CIoT optimization and user plane.

Specifically, this feature adds support for:

- Connected mode handovers for:
 - non-IP PDNs using S1U tunneling (without control plane CIoT optimization). This is possible only in WB-EUTRAN
- Idle mode handovers for:
 - non-IP PDNs with control plane CIoT optimization (S11U tunneling)
 - IP PDNs with control plane CIoT optimization (S11U Tunneling)
 - control plane only PDNs for either IP or non-IP type
- User plane switch from control plane CIoT optimization (S11U tunneling) to user plane (S1U) for UE during data transfer and idle mode handover.
- User plane switch from user plane (S1U) to control plane CIoT optimization (S11U tunneling) for UE during idle mode handover.
- Updating at CSGN if there is change in EPCO support during handovers (target MME supports the EPCO but not the source MME and vice versa)

The following use cases apply to this feature:

- This feature provides UE handover support and eliminates the need for the UE to reattach to get connectivity. The UE continues to remain connected, thereby reducing signaling, power, and consumption costs.

This feature enables the downlink data buffered in the core network to be delivered to the NB-IoT/CIoT supported device during handovers instead of requiring the core network to drop the data if the UE is forced to reattach.

Support for a local database to increase PGW capacity

MCC supports the use of a local database to handle subscriber transactions, thereby reducing the need to use MCC capacity. When there is no subscriber activity, subscribers are moved to the local database. When subscriber activity occurs, MCC moves the subscriber out of the database and handles the events and messages. After a period of inactivity, the subscriber is again moved to the database.

Configuration commands control when a subscriber is eligible to move to the database. The local database increases session capacity per-CPM VM. Operators use CLI commands and filters to display which subscribers are in memory or in the database.

Refer to [VM footprint requirements \(page 1-7\)](#) for details on VM sizing requirements to support 3M subscribers.

Limitations

This feature has the following limitations:

- Paging messages are not stored as part of compressed sessions and are therefore flushed. If the session has any paging data pending, then the session is not moved to the database. Also, if Fastpath crashes, the queued packet is lost.
- Extensive workflow debug counters are not stored (only partial information is retained)
- Higher packet loss occurs during WSM switchover
- GGSN calls are not moved to the database
- Only 1 Gi/Gn data packet is stored per session
- Support for SGW calls only is not supported

- Do not use the `zone default gateway subscriber clear imsi` or `zone default gateway subscriber count imsi` commands to clear subscribers when the scaling feature is enabled. Instead, use the `subscriber clear` and `subscriber count` commands.
- If the subscriber is in the database, then the `show sub pdn-session <pdn-session-id>` and `show sub pdn-bearer <pdn-session-id> <ebi>` commands will not display any data statistics where the PDN session ID belongs to the SGW. The MCC correctly displays all other details.

CLI Commands

Since the scaling feature works upon subscriber inactivity, the inactivity-timer-duration must be configured for the subscriber.

The inactivity-time-duration enables Operators to detect and potentially take action on subscribers that are inactive. Inactive is defined as no user data or control messages in the uplink or downlink direction for all PDN sessions and all bearers.

- SGWs – The inactivity-timer-duration is defined in the gateway-common profile.
- PGWs and GGSNs – The inactivity-timer-duration is defined in the gateway apn object if the value is non-zero, otherwise the value is defined in the gateway-common profile.

The subscriber/PDN session is marked as Idle or Active as follows:

- Idle – If no activity occurs within the configured inactivity period (in seconds), the subscriber/PDN session is declared Idle.
- Active – When a subscriber first connects to a GW, the timer is set and the subscriber/PDN session state is set to Active. In addition, if activity is detected on a subscriber/PDN session that had previously been declared Idle, the session is declared Active.

Note: The value defined in the gateway apn object takes precedence if the value is non-zero. Otherwise, MCC uses the value defined in the gateway-common profile.

Use one of the following methods to configure the inactivity-timer-duration in seconds. A value of 0 indicates not specified.

- Configure the inactivity-timer-duration at the APN level. The APN is added in the gateway using the apn-list.

```
an3000-mcm-slot7cpu1 config# zone default gateway apn consumer.att
inactivity-timer-duration                                     40
```

```
>>Snip<<
```

- Configure the inactivity-timer-duration in the gateway common profile using the profile in the gateway. The example is for a PDN gateway. You can use the same procedure for an SGW.

```
an3000-mcm-slot7cpu1(config-zone-default)# gateway profile
gateway-common default-profile
an3000-mcm-slot7cpu1(config-gateway-common-default-profile)#
an3000-mcm-slot7cpu1(config-gateway-common-default-profile)# show
full-configuration
zone default
gateway profile gateway-common default-profile
    gtp-echo-interval                                60000
>>>snip<<<
inactivity-timer-duration                            40
>>>snip<<<

an3000-mcm-slot7cpu1(config-pgw-MYPGW)# show full-configuration
zone default
gateway pgw MYPGW
    gateway-profile                                default-profile
>>>snip<<<
```

The following CLI command enables the scaling feature in fastpath. After adding the configuration, if the packet is received in fastpath and the session is not found, then packets without fastpath sessions are handled and the session is fetched from the database to memory. If the following CLI command is not issued, then the datapath drops the packet.

The default value is **false** indicating that IoT optimization is disabled in fastpath.

```
an3000-mcm-slot7cpu1(config)# ip-data-path stateless-wsm-mode true
```

The following command enables the threshold in the control path. This command applies to per-SGW and per-PGW processes. When this threshold is crossed, calls are moved to the database. If this value is set to 0, then sessions are not moved to the local database. If this value is set to 1, then all sessions are moved to the local database. For example, if the **push-to-db-threshold** is configured as **2**, then session 2 and subsequent sessions are moved to the database.

If any of the following conditions exist, the session is not moved to the database:

SGW

- Session is in extended paging state
- Subscriber is created for indirect tunnel
- Subscriber has multiple PDNs.

- Sessions without dedicated bearers
- Subscriber is not yet inactive

PGW

- Subscriber has dedicated bearers
- Subscriber session has any of the Gx/Gy re-joining timers running
- Subscriber session has active procedures running
- Subscriber has allocated address using DHCP
- Subscriber is not yet inactive

For example:

```
an3000-mcm-slot7cpu1(config)# zone default gateway sgw MYSGW
an3000-mcm-slot7cpu1(config-sgw-MYSGW)# push-to-db-threshold 2

an3000-mcm-slot7cpu1(config-zone-default)# gateway pgw MYPGW
an3000-mcm-slot7cpu1(config-pgw-MYPGW)# push-to-db-threshold 2
```

Note: This limit is per-PDN and per-SGW task. For example, if the number is *X*, it indicates that *X-1* sessions are kept in memory. After sessions are added, additional sessions are moved to the database.

Performance Statistics

The following performance statistics pertain to this feature.

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance sgw MYSGW | include database
num-sessios-in-local-database
1
an3000-mcm-slot7cpu1#

an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level call-performance pgw MYPGW | include
num-sessions-in-local-database
<<snip>>
num-sessions-in-local-database
1
<<snip>>

an3000-mcm-slot7cpu1#
```

```
an3000-mcm-slot7cpu1# show subscriber summary
subscriber show-summary
```

```
SUB
PDN
SUBSCRIBER SUBSCRIBER UPTIME MEMORY
SESSION GATEWAY GATEWAY
ACTIVITY LOCATION SUBSCRIBER IN MAC STORAGE
ID IMSI NAME TYPE MEID MSISDN
APN SERVICE NAME STATE STATE NAI SECONDS
ADDRESS STATE
-----
131073 17567547897024 MYSGW sgw 917567665558024
17567547897024 consumer.att SGW_24_1_1_7 active visiting
3 in-memory
167936 17567547897024 MYPGW pgw 917567665558024
17567547897024 consumer.att PDNGW_24_1_1_6 active visiting
3 in-memory
```

Show, Clear, and Count Commands

The following show commands pertain to this feature:

Note: The same database-type filter is available in the following commands:

```
subscriber clear database-type <in-memory|local-db>
subscriber clear-local database-type <in-memory|local-db>
subscriber clear database-type <in-memory|local-db>
```

```
aan3000-mcm-slot7cpu1# show subscriber count database-type in-memory
subscriber show-count
```

```
CLUSTER
ID COUNT
-----
44 16
```

```
an3000-mcm-slot7cpu1# show subscriber count database-type local-db
subscriber show-count
```

```
CLUSTER
ID COUNT
-----
44 4
```

```
an3000-mcm-slot7cpu1# show subscriber summary database-type in-memory
subscriber show-summary
```

```
SUB
PDN
SUBSCRIBER SUBSCRIBER UPTIME MEMORY
```

Features of C-SGN function

```
SESSION          GATEWAY GATEWAY
ACTIVITY         LOCATION SUBSCRIBER IN      MAC      STORAGE
ID      IMSI      NAME      TYPE      MEID      MSISDN      APN
SERVICE NAME     STATE      STATE      NAI      SECONDS  ADDRESS  STATE
-----
106497  17567547897024  MYSGW    sgw      917567665558024  17567547897024
consumer.att SGW_24_1_1_0  active    visiting      11
in-memory
147456  17567547897024  MYPGW    pgw      917567665558024  17567547897024
consumer.att PDNGW_24_1_1_0 active    visiting      11
in-memory

an3000-mcm-slot7cpu1# show subscriber summary database-type local-db
subscriber show-summary

SUB
PDN
SUBSCRIBER SUBSCRIBER          UPTIME          MEMORY
SESSION          GATEWAY GATEWAY
ACTIVITY         LOCATION SUBSCRIBER IN      MAC      STORAGE
ID      IMSI      NAME      TYPE      MEID      MSISDN      APN
SERVICE NAME     STATE      STATE      NAI      SECONDS  ADDRESS  STATE
-----
106497  17567547897024  MYSGW    sgw      917567665558024  17567547897024
consumer.att SGW_24_1_1_0  idle      visiting      61
local-db
147456  17567547897024  MYPGW    pgw      917567665558024  17567547897024
consumer.att PDNGW_24_1_1_0 idle      visiting      61
local-db
```

Procedure-duration Statistics

The following procedure-duration statistics pertain to this feature:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
procedure-duration saegw-sgw
gateway statistics procedure-duration saegw-sgw
```

		DURATION	
NAME	PROCEDURE NAME	IN MS	COUNT
MYSGW	create-session	100-149ms	1
MYSGW	move-from-db	0-49ms	1
MYSGW	modify-bearer-request	50-99ms	1
MYSGW	s1-release	50-99ms	1


```
an3000-mcm-slot7cpu1# show zone default gateway statistics
procedure-duration saegw-pgw
gateway statistics procedure-duration saegw-pgw
```

		DURATION	
NAME	PROCEDURE NAME	IN MS	COUNT

MYPGW	create-session	0-49ms	6
MYPGW	create-session	100-149ms	1
MYPGW	delete-session	0-49ms	1
MYPGW	get-from-db	0-49ms	36
MYPGW	modify-bearer-request	0-49ms	41
MYPGW	move-to-db	0-49ms	39
MYPGW	network-initiated-deactivate-subscriber	0-49ms	3
MYPGW	network-initiated-deactivate-subscriber	50-99ms	1
MYPGW	network-initiated-deactivate-subscriber	>= 5000ms	1
MYPGW	pending-local-close-subscriber	0-49ms	2

eDRX control message rejection handling at PGW

The MME rejects control messages (CBReq/UBReq) with cause code 123 (UE is temporarily not reachable due to power saving) if the UE is in power saving mode. The PGW reattempts the messages when the MME informs the UE of its availability.

This feature implementation is compliant with Release 13 GTPV2 specification, September, 2016.

Cause code 123 indicates that the UE is temporarily not reachable due to power saving. It is used by the MME/SGSN in the Create/Update Bearer Response message to reject the corresponding network-initiated procedures for a Delay Tolerant PDN connection and also request the PGW to hold the network-initiated procedure until it receives the subsequent Modify Bearer Request message with the UASI flag, indicating that the UE is available for end-to-end signaling.

***Note:** This feature is not supported between the MCC and the Affirmed Networks MME in this release.*

Configuration

Configure this object as follows:

To configure the delay-tolerant-connection-indication (DTCI) option in the APN:

```
an3000-mcm-slot7cpu1(config)# zone default
an3000-mcm-slot7cpu1(config-zone-default)# gateway apn consumer.att
```

Features of C-SGN function

```
an3000-mcm-slot7cpu1(config-apn-consumer.att)# delay?
Possible completions:

delay-tolerant-connection-indication-support  Enable/Disable DTCI for
the APN

an3000-mcm-slot7cpu1(config-apn-consumer.att)#
delay-tolerant-connection-indication-support ?
Possible completions:

<Enabled/Disabled>[disabled]

an3000-mcm-slot7cpu1(config-apn-consumer.att)#
delay-tolerant-connection-indication-support enabled
```

Show Commands

The PDN session displays the following status if the DTCI is enabled for this PDN. It also shows if any buffered update bearer, create bearer, or packet must be resent.

```
an3000-mcm-slot7cpu1# show subscriber pdn-session 167936 | include
delay    <<PGW

delay-tolerant-pdn-connection-enabled      true

an3000-mcm-slot7cpu1#

an3000-mcm-slot7cpu1# show subscriber pdn-session 75777 | include
delay    <<SGW

delay-tolerant-pdn-connection-enabled      true

an3000-mcm-slot7cpu1#

an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level procedure sgw | include PSM

update-bearer-reject-due-to-UE-in-PSM      0

nw-initiated-create-bearer-reject-due-to-UE-in-PSM 1

an3000-mcm-slot7cpu1#

an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level procedure saegw-sgw | include PSM

update-bearer-reject-due-to-UE-in-PSM      0

nw-initiated-create-bearer-reject-due-to-UE-in-PSM 1

an3000-mcm-slot7cpu1#
```

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level procedure saegw-pgw | include PSM

nw-initiated-modify-qos-reject-due-to-UE-in-PSM      0

nw-initiated-modify-bearer-reject-due-to-UE-in-PSM    0

nw-initiated-activate-bearer-reject-due-to-UE-in-PSM 1

an3000-mcm-slot7cpu1#

an3000-mcm-slot7cpu1# show zone default gateway statistics
cluster-level procedure pgw | include PSM

nw-initiated-modify-qos-reject-due-to-UE-in-PSM      0

nw-initiated-modify-bearer-reject-due-to-UE-in-PSM    0

nw-initiated-activate-bearer-reject-due-to-UE-in-PSM 1

an3000-mcm-slot7cpu1#
```

Logs

This feature outputs the following logs:

Enabling DTCI for the call on the PGW:

```
2017-03-31T00:18:25.822871747-0400 Sev=8 [Pgw] [GwCommon] [ID=23]
[0001180100011fe0] [07341 16069] [PDNGW_24_1_1_0_PDNGW_0] "[GW:MYPGW
IMSI:17567547897024 MEI:917567665558024 APN:consumer.att BearerID:5
State:PgwPdnConnCreatingState ProcedureID:0 SessionID:0x29005
Type:gtpV2]: Enable delay tolerant connection indication in CSRsp
```

Reception of UBRsp/CBRsp with cause code 123:

```
2017-03-31T00:19:05.856505800-0400 Sev=8 [Pgw] [GwCommon] [ID=23]
[0001180100011fe0] [07341 16069] [PDNGW_24_1_1_0_PDNGW_0] "[GW:MYPGW
IMSI:17567547897024 MEI:917567665558024 APN:consumer.att BearerID:5
State:PgwPdnConnActiveState ProcedureID:0 SessionID:0x29005
Type:gtpV2]: Create Bearer Response message received with casue code -
UE is temporarily not reachable due to power saving(123)
```

Reception of MBReq with UASI flag received:

```
2017-03-31T00:20:25.966633637-0400 Sev=8 [Pgw] [GwCommon] [ID=23]
[0001180100011fe0] [07341 16069] [PDNGW_24_1_1_0_PDNGW_0] "[GW:MYPGW
IMSI:17567547897024 MEI:917567665558024 APN:consumer.att BearerID:5
State:PgwPdnConnPSMState ProcedureID:0 SessionID:0x29005 Type:gtpV2]:
UE available for Signalling indication received in MBReq
```

Scheduling the procedures from pending list:

```
2017-03-31T00:20:25.981658936-0400 Sev=8 [Pgw] [GwCommon] [ID=03]
[0001180100011fe0] [07341 16069] [PDNGW_24_1_1_0_PDNGW_0] "[GW:MYPGW
IMSI:17567547897024 MEI:917567665558024 APN:consumer.att BearerID:5
State:PgwPdnConnPSMState/PdnGwSubCauseCode123State ProcedureID:17
SessionID:0x29005 Type:gtpV2]: UE Available for signalling event
received in MBReq. Schedule pending UBRsp/CBRsp
```

SGW forwarding UBRsp/CBRsp with cause code 123:

```
2017-03-31T00:19:05.843310812-0400 Sev=6 [Sgw] [GwCommon] [ID=09]
[0001180100011fea] [07248 14698] [SGW_24_1_1_0_SGW_0] "[GW:MYSGW
IMSI:17567547897024 MEI:917567665558024 APN:consumer.att BearerID:5
State:SgwPdnConnActiveState/SpcMmeCBRespState ProcedureID:10
SessionID:0x1280b Type:gtpV2]: Received Create Bearer Response with
Cause Code 123
```

Limitations

The following limitations apply to this feature:

- Edrx/PSM is not supported for GGSN call
- pcrf-send-crr-in-raa does not enforce sending charging rule report if call is DTCl enabled and also RAA is sent immediately without waiting for access-side signaling
- If PCRF requests a charging rule report but the access side signaling is not completed when UE in PSM, PGW does not send any charging rule report
- Pending procedures are not preserved across redundancy actions
- No enforcement on the maximum number of pending procedures
- Cause code of PSM is not handled in following cases
 - 3gpp to Non-3gpp handover cases
 - Non3gpp to 3gpp handover cases
 - cause code 123 not handled for PCO Update: UBRsp
- If UE is in PSM, the following actions are queued until the UE becomes available:
 - Gx rejoin
 - OCS rejoin
 - Profile switch
 - CRBN profile
 - Deferred rules processing

P-GW Congestion Protection for MTC use cases (SM-Back-off)

The PGW rejects the call if the APN is congested. The PGW sends the EPC (back-off timer) timer IE to the MME in the Create Session Response, so that the MME does not send calls during congestion. After receiving the Create Session Response, the MME may decide to redirect the call to a different PGW. The rejection cause code is set to APN Congested (113).

To enable this feature, set the apn-congestion-threshold admin-state to enable. By default, the APN state is **APN Not Congested**. The EPC timer value is fixed at 10 seconds.

Threshold detection occurs periodically, every 30 seconds. Therefore, there is a chance that a few calls might get accepted, even if the threshold crosses the configured value.

***Note:** This release supports critical-enter (crossing critical threshold) only. Threshold-type is preset to op and has no impact on feature behavior.*

The APN load is calculated based on number of sessions only. Only the APN received in the session Request is used to reject calls. The load on the mapped APNs is not considered.

If the Operator configures both the SGW and PGW on the same cluster, then only calls at the PGW are counted against the APN.

The following example configuration causes the gateway to reject the call with cause code 110 and an EPC timer value of 10 seconds, if the number of calls established on the APN is greater than 10. If the number of calls falls below 10, then the gateway starts accepting new calls for that APN.

```
an3000-mcm-slot7cpu1# config
Entering configuration mode terminal
an3000-mcm-slot7cpu1(config)# zone default
an3000-mcm-slot7cpu1(config-zone-default)# gateway
apn anp.name
an3000-mcm-slot7cpu1(config-apn-anp.name)# apn-congestion-threshold ?
Possible completions:
admin-state Enable/Disable the administrative
state of this threshold
critical-enter Crossing critical threshold
major-enter Crossing major threshold
minor-enter Crossing minor threshold
threshold-type
an3000-mcm-slot7cpu1(config-apn-anp.name)# apn-congestion-threshold
```

```
admin-state enabled critical-enter 10
an3000-mcm-slot7cpu1(config-apn-anp.name)# commit
Commit complete.
an3000-mcm-slot7cpu1# config
Entering configuration mode terminal
an3000-mcm-slot7cpu1(config)# zone default
an3000-mcm-slot7cpu1(config-zone-default)# gateway apn anp.name
an3000-mcm-slot7cpu1(config-apn-anp.name)# show full-configuration
zone default
gateway apn anp.name
apn-restriction 1
<<snip>
apn-congestion-threshold critical-enter 10
apn-congestion-threshold admin-state enabled
apn-congestion-threshold threshold-type op
!
```

Support for SGW extended buffering

This release supports extended buffering on NBIoT devices that are in power saving mode. Previously, MCC kept all the paging buffers in the platform buffers only. Using extended buffering in this release, the page queue maintains only heap buffers. These buffers are classified under HeapPaging buffer type. This release adds support for the new **max-heap-paging-size** object to control the maximum heap memory that can be used in WSM, based on WSM memory and Operator requirements. For example:

```
an3000-mcm-slot7cpu1(config)# ip-data-path max-heap-paging-size ?
Possible completions:
  <unsignedInt, 10 .. 800000>[203]
```

MCC uses the existing **max-downlink-pkts-to-buffer** configuration of page buffers when the extended buffering feature is not used.

This release adds support for the **maxcap-downlink-pkts-to-buffer** object to cap the maximum number of page buffers for any given session, when using the extended buffering feature. For example, if MME responds with more than the local configuration, MCC uses the local configuration value. This object captures the maximum downlink packets to buffer in case of paging, when the UE is in power saving mode if MCC receives a large DLBufferingPacketCount value from the MME. This object aids in protecting the MCC paging buffers in case of a wrong configuration on the MMEs.

Configure this object as follows:

```
an3000-mcm-slot7cpu1(config)# zone default gateway sgw NYsgw
paging-profile
max-downlink-pkts-to-buffer          Maximum number of downlink data
```

```
packets to buffer per bearer when UE is being paged, when Number of
Page Packets are not received from MME
maxcap-downlink-pkts-to-buffer    Maximum number of downlink data
packets to Cap when Number of Page Packets are received from MME
an3000-mcm-slot7cpu1(config)# zone default gateway sgw NYsgw
paging-profile maxcap-downlink-pkts-to-buffer ?
Possible completions:
<unsignedInt, 1 .. 20>[5]
```

Support for non-IP data via tunneling (non-IP PDN type)

The C-SGN supports non-IP data services via tunneling on the SGi.

Previously, the LTE specification supported the IP PDN type only. Release 13 allows a subscriber to use a non-IP PDN type as part of the NB-IoT changes. The PGW allocates IPv4/6 address for a subscriber but does not pass this information to the SGW/UE as part of the Create Session Response message.

Operators must configure the local IPv4/6 pool. Based on the APN configuration, the IPv4/6 address is allocated for non-IP PDN type subscribers (ipv4/6 is not a requirement).

The *GTP v2 Create Session Request* message contains the PDN Type as non-IP and PAA contains the pdn-type field.

The PGW maintains the PDN type as non-IP but the UE IP address is allocated as v4/v6/v4v6 depending on the APN configuration. The SGW can have IP and non-IP PDN types for different PDN connections for a given subscriber.

Gi Side UDP Tunneling for Non-IP

The 3GPP LTE specification allows non-IP as a PDN type, in addition to the IP PDN type. A Gi network is typically an IP network, therefore, a tunneling mechanism must be used to transport non-IP packets. The 3GPP specification defines a UDP tunnel as the preferred mechanism to transfer non-IP packets. A UDP tunnel is created between the GW and Application Server (AS) as follows:

- Source IP Address (not configured but uses UE IP address).
- Source Port (configured as part of tunnel).
- Destination IP address (Application Server IP address configured as part of tunnel).
- Destination Port (Application Server port configured as part of tunnel).
- Tunnel object is already defined under network context and modified to make it a UDP tunnel.

Workflow control-profile uses existing tunnel attribute as UDP tunnel (similar to GRE tunnel).

Enhanced the existing tunnel configuration to support a non-IP UDP tunnel. For example:

```
an3000-mcm-slot7cpu1(config)# network-context NYC tunnel UDPTunnel1
type UDP remote-ip-address 1.2.3.4 source-port 50000 destination-port
52000
an3000-mcm-slot7cpu1# show running-config network-context tunnel
an3000-mcm-slot7cpu1# show running-config network-context tunnel
network-context NYC
tunnel UDPTunnel1
    type                UDP
    remote-ip-address    1.2.3.4
    source-port          50000
    destination-port     52000
```

After creating a UDP tunnel, add it to the workflow control-profile. For example:

```
an3000-mcm-slot7cpu1(config)# workflow control-profile wfc1
default-tunnel-configuration UDP?
```

Possible completions:

```
[UDP2]  UDP1  UDP2  UDPTunnel1
```

```
an3000-mcm-slot7cpu1(config)# workflow control-profile wfc1
default-tunnel-configuration UDPTunnel1
```

Protocol configuration options for MTU size of non-IP data over NAS

This release supports protocol configuration options (PCO) for maximum transmission unit (MTU) size of the non-IP small data over NAS.

PCO vs ePCO

Based on TS 24.301:

- The GW receives either PCO or ePCO. If the GW receives both PCO and ePCO, all PCO configuration response pertain to the ePCO attribute.
- If the GW receives PCO with new NB-IoT-related attributes, it does not reject the PCO but responds with all the configured information in the PCO attribute.
- If the GW receives ePCO with old and new attributes, the GW responds with the configured information in the ePCO attribute.

As the non-IP data is passed transparently by the C-SGN and NB-PGW, it configures the MTU size for non-IP traffic.

Added support for Extended PCO to augment the current PCO. Extended PCI is indicated in the `Indication Flag` as part of Create Session Req/ Modify Bearer Req messages.

Non-IP Link Level MTU Configuration

Added support for non-IP link MTU to the zone gateway APN configuration. Operators can configure the non-IP link MTU size (between 128 and 1600) for non-roaming and roaming PDN connection to be sent in the extended PCO IE. For example:

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn vf.consumer
epco-nonip-link-mtu-size-
```

Possible completions:

```
epco-nonip-link-mtu-size-non-roaming    Non IP Link MTU Size for a
Non-Roaming PDN Connection to be sent in the Extended PCO IE
```

```
epco-nonip-link-mtu-size-roaming        Non IP Link MTU Size for a
Roaming PDN Connection to be sent in the Extended PCO IE
```

Added a non-IP level MTU configuration (similar to existing IPv4 link level) at the APN level.

```
an3000-mcm-slot7cpu1(config)# zone default gateway apn consumer.att
pco-nonip-link-mtu-size-non-roaming 1384
```

NBIOT PGW support for new PDN type

The C-SGN supports PDN connection establishment message and establishing IP Bearer Service.

For IP Data Type

The PDN type indicates type of IP address (that is, IPv4, IPv6) to be allocated. The C-SGN allocates an IP address based on the PDN type in the Attach Request. No NAS Session Management signaling is needed. In the roaming case, the C-SGN sends a Create Session Request (or new control message) to the home C-SGN/PGW indicating this is CIoT attach and Data Type. the C-SGN allocates an IP address based on the PDN type in the Attach Request.

For Non-IP Data Type

The C-SGN does not provide any IP address to the UE. The PGW part of the C-SGN allocates an IP address for use across SGi. Based on the configuration, the C-SGN may establish a direct forwarding path, for example, point-to-point tunnel on a per-UE per-PDN basis towards the AS. Alternatively, based on the HSS provided subscriber data, the C-SGN may be required to route small data via SCEF.

The C-SGN supports attach procedures for SMS only, that is, attach only for sending and receiving SMS without any PDN connection for IP (or non IP) data.

Support for the NBIOT RAT Type

The C-SGN supports the **nbiot-eutran** RAT (Radio Access Technology) type, as specified by Release 13, 3GPP specification. This new RAT type is introduced for massive Machine Type Communications (MTC) applications with cost-effective connectivity with simplified low-cost devices and minimizes battery usage. For example:

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <SA> key
<AK> rat-type
possible completions:
eutran  gan  geran  hspaEvolution  nbiot-eutran  utran  virtual  wlan
```

The show subscriber summary filter is enhanced with **nbiot-eutran** rat type. For example:

```
an3000-mcm-slot7cpu1# show subscriber summary rat-type nbiot-eutran
subscriber show-summary
```

SUBSCRIBER	SUBSCRIBER	GATEWAY		UPTIME			
PDN	ACTIVITY	LOCATION	SUBSCRIBER	IN	MAC		
SESSION ID	IMSI	NAME	TYPE	MEID	MSISDN	APN	
SERVICE NAME	STATE	STATE		NAI	SECONDS	ADDRESS	
2148909061	12343222	NYsgw	sgw	917567662558800	917567662558800		
consumer.att	SGW_12_1_1_0	active	home			88469	

Show subscriber pdn-session command output for RAT Type incase of NBIOT would be as follows

```
an3000-mcm-slot7cpu1# show subscriber pdn-session 2149531664
subscriber pdn-session 2149531664
default-bearer-id 5
default-logical-bearer-id 0
ue-v4-ip-address 10.48.12.15
prior-apn-name consumer.att
access-in-control-teid-interface-type s5s8-pgw-gtpc
..
uplink-hairpin-policy-discard-packets 0
uplink-hairpin-error-discard-packets 0
uplink-tunneled-packets 0
rat-type NBIOT-EUTRAN
pdn-type ipv4
workflow-subscriber-analyzer anlyz1
l2tp-tunnel-endpoint ::
l2tp-tunnel-id 0
```

Updated the following additional objects to support the **nbiot-eutran** RAT type:

- Subscriber summary filter
- External interfaces such as RADIUS, OFCS, OCS, GTPP, and PCRF

- Diameter-related RAT type attributes for PCRF, OCS, and OFCS.
- EDR protobuffs to reflect in EDRs

Note: In Release 13, 3GPP specification, the existing LTE RAT type **E-UTRAN** is renamed to **WIDE BAND EUTRAN**. Affirmed Networks is not renaming the RAT type in the CLI but Operators should assume that **EUTRAN** now refers to **WIDE BAND EUTRAN** in the Affirmed Networks CLI.

Added support for ip and nonip PDN types.

Added the ip and nonip key PDN types as part of workflow subscriber-analyzer keys. Using this key, subscribers with IP and non-IP PDN types can use the same APN but different services based on the workflow control profile.

For example:

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer only3 key
ApnPdpType pdn-type
Possible completions:
ip  nonip
```

CLI command to allow account administrators to change the default SFTP directory of user accounts

CLI users (except users with ready-only access), are permitted to change their own default SFTP directory using the `user-access change-sftp-dir` command.

This feature adds a new CLI command to allow account administrators (admin group members and the default CALEA user) to change the default SFTP directory of user accounts for which they have administrator privileges, including read-only users. Administrators must set the SFTP directory to either the user's "home" (default) directory or a subdirectory under the `/public/*` directory.

```
an3000-mcm-slot7cpu1# user-access reset-sftp-dir user <username>
```

The system prompts:

```
Enter sftp directory:
```

Enter the path to set the default SFTP directory.

Note: Configuring an SFTP directory for a read-only user enables SFTP/SCP for that user. Typically, read-only users whose SFTP directory is set to "home" are not permitted to use SFTP/SCP.

Index

C

CALEA [2-1](#)
 log user commands [2-2](#)
CDR
 GTPP CDRs [2-10](#)
 PGW CDRs [2-10](#)
 related NB-IOT AVPs [2-8](#)
C-SGN [1-1](#)
 overview of C-SGN [1-1](#)

D

DoNAS [1-1](#)

E

EUTRAN [2-12](#)

G

Ga interface [2-10](#)
Gx Interface [2-9](#)
Gy Interface [2-8](#)
Gz/Rf Interface [2-9](#)

L

Lawful Intercept [2-1](#), [2-2](#)

M

Management Configuration Module [1-7](#)
MCC [1-ix](#), [1-xi](#)

N

Narrowband-EUTRAN [2-12](#)

P

PDN [1-2](#)
Personality Types [1-7](#)

S

Subscriber Services Module [1-7](#)

T

TACACS
 enable/disable TCAS AAA [2-1](#)

V

VM Types [1-7](#)

W

WB-E-UTRAN [2-12](#)

Wideband-EUTRAN [2-12](#)